

RAMKUMAR U

Full Stack Developer | IoT & Industrial Automation Software Engineer

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SUMMARY

Full Stack Developer with 2+ years building production web, desktop, and mobile applications using React (TypeScript), Node.js, Python (Flask/FastAPI), and MySQL, plus specialized expertise in IoT/industrial automation and applied AI (RAG-based LLM assistants). Skilled at owning a product end-to-end — frontend UI, backend APIs, databases, third-party integrations (WhatsApp, IndiaMART), and AWS/Firebase deployment. Strong problem-solver focused on scalable, secure, user-centric software.

CORE TECHNICAL SKILLS

- Frontend:** React.js, Redux, Context API (Global State Management), TypeScript, Sass/SCSS, MUI, Bootstrap 5
- Backend:** Node.js, Express.js, Python (Flask, FastAPI), REST API design, JWT Auth
- Database:** MySQL
- Cloud & Deployment:** AWS EC2 (provisioning, security groups, firewall, reverse proxy, domain/SSL setup), AWS Amplify, Firebase Hosting, Windows IIS deployment
- Desktop & Mobile:** Electron (auto-launch, auto-updater, logging) for Windows desktop apps; Capacitor for Android (APK) apps
- IoT & Industrial Automation (Specialization):** PLC integration, Modbus RTU (RS485/RS232) & TCP/IP communication, RFID & barcode workflows, IP camera / OpenCV vision, receipt & barcode printer integration, pymodbus, SQLAlchemy
- AI & Integrations:** RAG-based LLM knowledge assistants, Meta WhatsApp Business API, IndiaMART API, third-party lead/CRM integrations
- Tools, Methodologies & Practices:** Git/GitHub, Postman, Agile/Scrum, Jira

EXPERIENCE

- Full Stack Developer — Bharath Vending Corporation** Feb 2024 – Present
Full Stack, IoT & Automation Software Developer
- End-to-end ownership:** designed and delivered web dashboards, Windows desktop apps, and Android apps to control, monitor, and manage industrial machines in real time.
 - Backend & architecture:** built backend architecture, MySQL database design, and REST APIs powering all machine dashboards and apps.
 - Cloud deployment:** managed production servers on AWS EC2, including security groups, firewall configuration, reverse proxy, and domain setup.
 - Hardware integration:** connected PLCs, RTU (RS485/RS232) and TCP/IP devices, RFID readers, barcode scanners, IP cameras, and printers for secure, real-time automation.
- Full Stack Developer — Seilax Software Systems Pvt. Ltd.** Sep 2023 – Feb 2024
Innovative Web Solutions
- Shipped production apps:** Company Profile Website, Real Estate Portal, and Mall Leasing & Management System using React, Node.js, and Firebase.
 - Frontend architecture:** built scalable, reusable React component architecture with centralized API services and global state management (MUI, Bootstrap, Framer Motion, Swiper, Axios, React Toastify, XLSX).
 - Backend APIs:** developed secure RESTful APIs with Node.js/Express (Axios, Body Parser, CORS, dotenv, Nodemon).
 - Live projects:** [www.seilaxsoftware.com](#) | [portal.seilaxsoftware.com](#)

PROJECTS

- ERP Software with AI Assistant — Bharath Vending Corporation**
- Designed and built a fully automated, company-wide ERP system from scratch covering operations, inventory, and lead management.
 - Integrated a RAG-based LLM AI assistant trained on internal company knowledge to answer employee/business queries instantly.
 - Built Meta WhatsApp Business API integration for automated customer communication directly from the ERP.
 - Connected IndiaMART API to automatically capture and sync inbound sales leads into the ERP in real time.
- Caterpillar Decal Vending Machine Software**
- End-to-end IoT vending solution with a web dashboard and desktop automation app — RFID & barcode-based automatic decal dispensing, role-based access, and real-time machine monitoring.
- Milk Weight Monitoring & Anti-Theft System**
- Dashboard and Android app to track vehicle and BMC-wise milk weights in real time, generate Bluetooth-printer receipts, and prevent theft via remote weight validation.
- Advanced Vertical Storage System (ASRS)**
- Automated storage and retrieval system with role-based access, inventory tracking, low-stock email alerts, and desktop-controlled industrial automation.
- Orange Juice Automation Machine**
- Fully automated juice vending system with desktop machine control, stock management, and PhonePe payment integration for cashless ordering and dispensing.
- Part Validation & Quality Inspection Software**
- Industrial part validation system using PLC integration and IP cameras — QR-based part tracking and automated visual inspection for quality assurance.

EDUCATION

- BE in Mechatronics Engineering** 2023
M.A.M School of Engineering, Trichy, India